

The horizontal hinge, called the flapping hinge, allows the blade to move up and down. This movement is called flapping and is designed to compensate for dissymmetry of lift. The flapping hinge may be located at varying distances from the rotor hub, and there may be more than one hinge.

The vertical hinge, called the lead-lag or drag hinge, allows the blade to move back and forth. This movement is called lead-lag, dragging, or hunting. Dampers are usually used to prevent excess back and forth movement around the drag hinge. The purpose of the drag hinge and dampers is to compensate for the acceleration and deceleration caused by Coriolis Effect.

Each blade can also be feathered, that is, rotated around its spanwise axis. Feathering the blade means changing the pitch angle of the blade. By changing the pitch angle of the blades you can control the thrust and direction of the main rotor disc.

SEMIRIGID ROTOR SYSTEM

A semirigid rotor system is usually composed of two blades which are rigidly mounted to the main rotor hub. The main rotor hub is free to tilt with respect to the main rotor shaft on what is known as a teetering hinge. This allows the blades to flap together as a unit. As one blade flaps up, the other flaps down. Since there is no vertical drag hinge, lead-lag forces are absorbed through blade bending. [Figure 5-6]

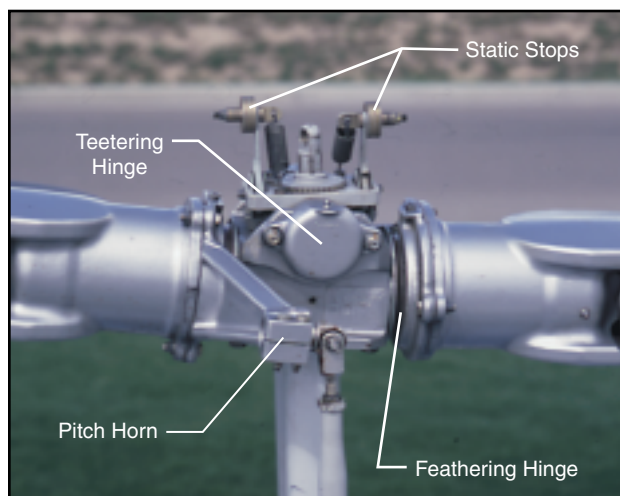


Figure 5-6. On a semirigid rotor system, a teetering hinge allows the rotor hub and blades to flap as a unit. A static flapping stop located above the hub prevents excess rocking when the blades are stopped. As the blades begin to turn, centrifugal force pulls the static stops out of the way.

RIGID ROTOR SYSTEM

In a rigid rotor system, the blades, hub, and mast are rigid with respect to each other. There are no vertical or horizontal hinges so the blades cannot flap or drag, but

they can be feathered. Flapping and lead/lag forces are absorbed by blade bending.

COMBINATION ROTOR SYSTEMS

Modern rotor systems may use the combined principles of the rotor systems mentioned above. Some rotor hubs incorporate a flexible hub, which allows for blade bending (flexing) without the need for bearings or hinges. These systems, called flextures, are usually constructed from composite material. Elastomeric bearings may also be used in place of conventional roller bearings. Elastomeric bearings are bearings constructed from a rubber type material and have limited movement that is perfectly suited for helicopter applications. Flextures and elastomeric bearings require no lubrication and, therefore, require less maintenance. They also absorb vibration, which means less fatigue and longer service life for the helicopter components. [Figure 5-7]



Figure 5-7. Rotor systems, such as Eurocopter's Starflex or Bell's soft-in-plane, use composite material and elastomeric bearings to reduce complexity and maintenance and, thereby, increase reliability.

SWASH PLATE ASSEMBLY

The purpose of the swash plate is to transmit control inputs from the collective and cyclic controls to the main rotor blades. It consists of two main parts: the stationary